Matthew Cranny

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Systems engineer developing automated hardware tests, custom electronics, and embedded C firmware. Experience spans board qualification, instrumentation, firmware deployment, aerospace hardware assembly, and ASIC design and verification.

EXPERIENCE

PDF Solutions March 2026–Present
Systems Engineer *San Jose, CA*

- Own qualification, calibration, and production acceptance across six board families using Python, PyQt6, bench instruments, SCPI, GPIB, and NI-DAQmx for electrical, analog, thermal, telemetry, and firmware testing.
- Developed a custom selector for four electrostatic correction boards; wrote C firmware with SCPI-style commands over RS-485 to route measurement outputs and Ethernet control for the active board.
- Built a system of record for over 1,000 items using Rust microservices and PostgreSQL, integrating frontend and backend workflows to track boards, serialized parts, installations, software, firmware, and configurations.
- Transitioned board testing to a custom OpenTAP front end and OpenHTF back end, standardizing test workflows and interfaces to improve reliability and scalability.
- Investigated power-up IP assignment and connectivity failures surfaced by faster automated testing; used AMD Vivado and register-level analysis to diagnose hardware and firmware faults.
- Coordinated overseas and field teams for firmware deployments, arranging scan comparisons before and after upgrades to verify equal or better performance, supplying updated hardware, and guiding remote installation.
- Characterized SEM settling with Python signal analysis and tuned inertia compensation, reducing time for oscillations to fall below 2 nanometers from 300 to 20 nanoseconds.

Amazon May 2025–July 2025
Area Manager Intern, ICQA *Turlock, CA*

- Built Python API and analysis pipelines for Inventory Control and Quality Assurance; migrated anomaly detection to AWS Lambda, EventBridge, and Secrets Manager, reducing estimated annual vendor transaction risk by $100,000.
- Automated labor reporting across fulfillment centers with VBA, eliminating over 120 hours of manual work each week.

Space Sciences Laboratory, UC Berkeley June 2024–May 2025
Aerospace Engineering Intern *Berkeley, CA*

- Modeled structural components in SolidWorks for LuSEE-Night, a NASA-funded lunar radio observatory; fabricated and assembled satellite hardware in cleanrooms under contamination-control protocols.
- Assembled high-voltage step-up converters and installed flight-hardware wiring harnesses and passive components.

SELECTED ENGINEERING PROJECTS

Programmable Protocol Emulator ASIC
Jane Street Protocol Emulator ASIC Competition

- Designed a dual-engine Verilog ASIC targeting 48 MHz on IHP CMOS5L, with 256 × 32 SRAM per engine, a shared 24-bit timebase, eight deadline-driven pin cells, and timestamped capture.
- Built a Python assembler and SPI loader with model-based RTL tests and formal proofs; ran V3 on an Artix-7 Cmod A7-35T and captured output timing with Vivado ILA.

Combat Robotics at Berkeley August 2023–May 2025
Technical Lead

- Built weapon and drive power systems integrating motor controllers, brushless motors, wiring harnesses, and battery management; designed packaging to protect electronics from shock and vibration.

EDUCATION

University of California, Berkeley May 2026
Bachelor of Arts in Physics
Coursework: Instrumentation Laboratory, Advanced Experimental Laboratory, Electromagnetism and Optics, Quantum Mechanics, Statistical and Thermal Physics

TECHNICAL SKILLS

Hardware and Test: Oscilloscopes, custom test hardware, PyVISA, AMD Vivado, SolidWorks, cleanroom assembly
Software: Python, C, Rust, Verilog, SQL, PostgreSQL, LabVIEW, pytest, cocotb, Icarus Verilog, Yosys, GitHub Actions, Linux